

ARG26 Spring Technical Report: Charging System

Nathaniel Su

NetID: nbs63

Subteam: Powertrain Peripherals

Team: Powertrain

Component: Charging System

Date: May 17, 2026

Table of Contents

Table of Contents.....	2
1. Introduction.....	3
1.1 Applicable FSAE Rules	3
2. Technical Overview	5
2.1 AC Input and J1772 Interface	5
2.2 Elcon Charger	5
2.3 High-Voltage Enclosure.....	6
2.4 Low-Voltage Control Enclosure	6
2.5 Charging Shutdown Circuit	6
3. History of Past Designs.....	8
3.1 BRUSA NLG513 Era	8
3.2 Transition to Elcon Charger (ARG24).....	8
3.3 ARG25 Charger	8
4. ARG26 Design.....	9
4.1 Overall Design Approach	9
4.2 Design Changes Since Fall 2025 Technical Report.....	10
4.3 High-Voltage Enclosure.....	10
4.4 Low-Voltage Control Enclosure	12
4.5 Charger Shutdown PCB.....	14
4.6 Harnessing and Connectors.....	16
4.7 Incomplete Aspects and Future Design Work	17
5. Manufacturing and Assembly	17
5.1 Enclosure Manufacturing.....	17
5.2 High-Voltage Connector Assembly.....	18
5.3 Elcon Charger Mounting.....	18
5.4 ARG27 Manufacturing Recommendations.....	18
6. Test Results.....	18
6.1 Completed Testing	18
6.2 Data Collection	19
6.3 Future Testing	20
7. References.....	20

1. Introduction

The ARG26 charging system is the off-car electromechanical assembly responsible for charging the tractive-system accumulator. It accepts AC power from EV supply equipment (EVSE) through the SAE J1772 interface, converts that AC power to regulated DC using the Elcon charger, and delivers charging current to the accumulator. Functionally, the charging system is a simplified off-car counterpart to the vehicle powertrain: it contains high-voltage distribution, low-voltage control, shutdown circuitry, safety interlocks, and measurement points required for safe tractive-system operation.

The charging system is necessary because the accumulator must be charged outside of the car in a controlled, rule-compliant manner. Although the charger does not directly improve vehicle performance during dynamic events, a reliable charging system is critical to competition readiness. A charger failure can prevent the accumulator from reaching full state of charge, delay accumulator technical inspection, or create unsafe high-voltage conditions. For ARG26, the primary design goals were reliability, rule compliance, clean harnessing, serviceability, and thorough documentation. These goals were prioritized over mass and packaging efficiency because the charging is an off-car system.

The charging system encompasses the following major elements: the Elcon charger, high-voltage charging leads, high-voltage distribution enclosure, low-voltage control enclosure, charger-specific shutdown PCB, discharge circuitry, J1772 interface circuitry, tractive-system measuring points (TSMPs), status indicators, and all external harnessing. The system is designed so that the charger output can only become live when correctly connected to the accumulator and when the charging shutdown circuit is closed.

1.1 Applicable FSAE Rules

The following 2026 FSAE rules govern the design, construction, and operation of the ARG26 charging system [1]:

Rule	Requirement
EV.2.1	Electrical System Form must describe the full charging system architecture.
EV.3.3.2	Maximum tractive-system voltage must not exceed 600 V DC.
EV.4.4	System must comply with grounded low-voltage rules.
EV.4.9	Tractive-system components must bear ISO 7010-W012 warning stickers.
EV.4.10.1–4	Tractive battery must be transported on a compliant hand cart and securely attached during transport.

EV.5.8.2 / EV.8.2.6	Two tractive-system measuring points (TSMPs) must be installed in the charger, connected to positive and negative charger output lines, and accessible during charging.
EV.6.3–6.5	Wiring, connections, and voltage separation must satisfy FSAE requirements.
EV.7.1.6	Grounding requirements must be satisfied.
EV.8.1.1	All charger and charging shutdown circuit functions must be demonstrated at Electrical Technical Inspection.
EV.8.1.2	Chargers are sealed after technical inspection approval.
EV.8.2.1	Charger must be galvanically isolated from AC input to DC output.
EV.8.2.2	Conductive charger housing must be connected to AC input earth ground.
EV.8.2.3	All charger connections must be isolated and covered.
EV.8.2.4	Charger connector must only become live when correctly connected to the tractive battery.
EV.8.2.5	High-voltage charging leads must be orange.
EV.8.2.7	Charger must include a compliant charger shutdown button.
EV.8.3.1–3.2	Charging shutdown circuit must include the shutdown button, BMS, and IMD; BMS and IMD paths must be independent and normally open.
EV.8.4.1–4.2	On a fault, shutdown circuit must open immediately, tractive system must discharge to low voltage within 5 seconds, charger must turn off, and must remain disabled until manually reset.

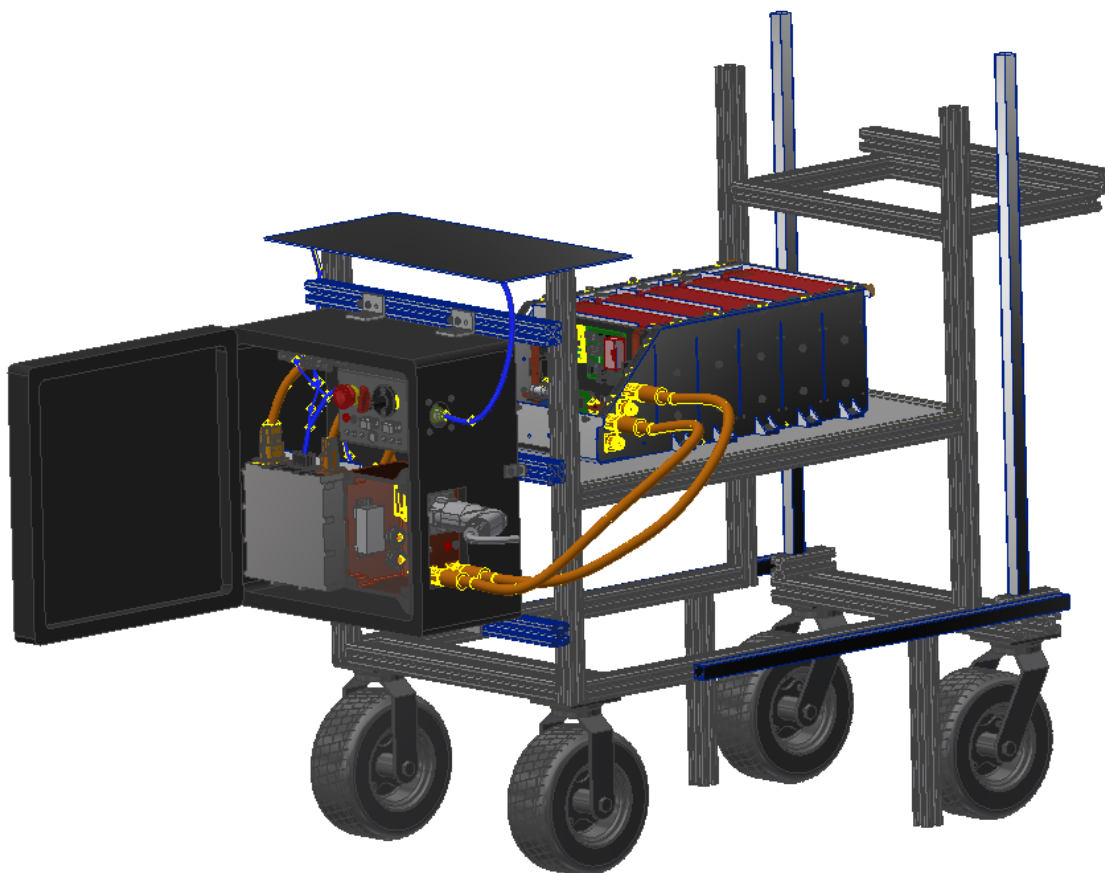


Figure 1: Full ARG26 charging system overview

2. Technical Overview

The charging system is divided into four major subsystems: the AC input and J1772 interface, the Elcon charger, the high-voltage (HV) enclosure, and the low-voltage (LV) control enclosure. These subsystems interact through high-voltage power lines, low-voltage control signals, CAN communication, and shared ground references. The overall system architecture is shown in Figure 2.

2.1 AC Input and J1772 Interface

The SAE J1772 standard defines the physical connector, pilot signal protocol, and proximity detection used for Level 1 and Level 2 AC charging. The J1772 pilot signal is a 1 kHz PWM waveform whose duty cycle encodes the maximum current available from the EVSE. The Active Vehicle Controller (AVC2.r) monitors the pilot and proximity signals and enables AC power delivery only when the connector is fully latched and the LV control system is operational. This prevents the charger input from becoming live before the system is in the correct operating state, satisfying EV.8.2.4. The AVC2.r handles the pilot state machine transitions (States A through D) required for standard-compliant charging initiation [3].

2.2 Elcon Charger

The Elcon charger is a power-factor-corrected AC-to-DC converter. It accepts single-phase AC input and produces a regulated DC output with programmable voltage and current limits set via CAN. The charger provides galvanic isolation between its AC input and DC output, satisfying EV.8.2.1. The Elcon has a separate low-voltage supply input for its internal control electronics. During operation, this supply is switched to an Elcon-generated 12 V rail once the charger is active, reducing drain on the external LV battery [2].

2.3 High-Voltage Enclosure

The HV enclosure contains all high-voltage distribution components associated with the charger output: the charger output connections, an in-line UL Class CC fuse, discharge circuitry, TSMPs connected to the positive and negative charger output lines, and SurLok Plus connectors for the accumulator interface. The TSMPs are shrouded two-pin connectors that allow voltage measurement without exposing the operator to bare high-voltage contacts, satisfying EV.5.8.2 and EV.8.2.6 [4].

The discharge board is a resistive circuit that brings the charger output to low-voltage levels within the five-second window required by EV.8.4.1 after the charging shutdown circuit opens. The discharge is activated automatically when the shutdown circuit opens by removing the signal that holds the discharge path off.

2.4 Low-Voltage Control Enclosure

The LV enclosure contains all user interface and control elements: the charger shutdown button, a manual reset button, source-selection switching between the external LV battery and the Elcon-supplied LV rail, status indicator LEDs, LV measuring points, mil-spec connectors, and the charger shutdown PCB. The enclosure is mounted to the top of the charging case so that the shutdown button and controls are accessible during charging and during Electrical Technical Inspection, as required by EV.8.1.1 and EV.8.2.7.

2.5 Charging Shutdown Circuit

The charging shutdown circuit is a normally-open series circuit that, when closed, enables the charger to deliver current to the accumulator. The circuit includes the charger shutdown button (normally-closed, push-to-open), the BMS shutdown output (normally open, closes when the battery is within safe limits), and the IMD shutdown output (normally open, closes when isolation is acceptable). Any single fault — an operator pressing the shutdown button, a BMS cell overvoltage, or an IMD isolation fault — immediately opens the circuit and disables charging, consistent with EV.8.3.1–3.2 and EV.8.4.1–4.2. The shutdown circuit topology is shown in Figure 4.

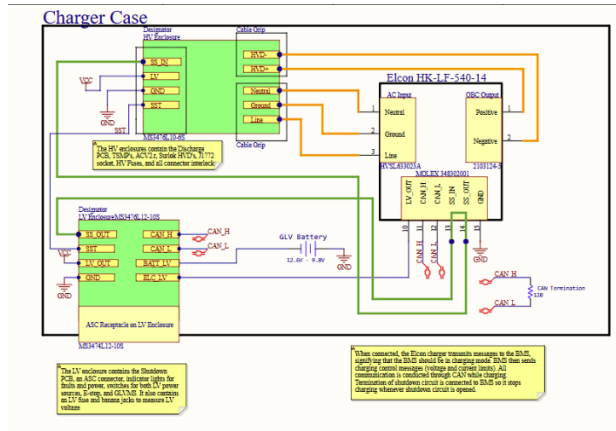


Figure 2: Top-level charging

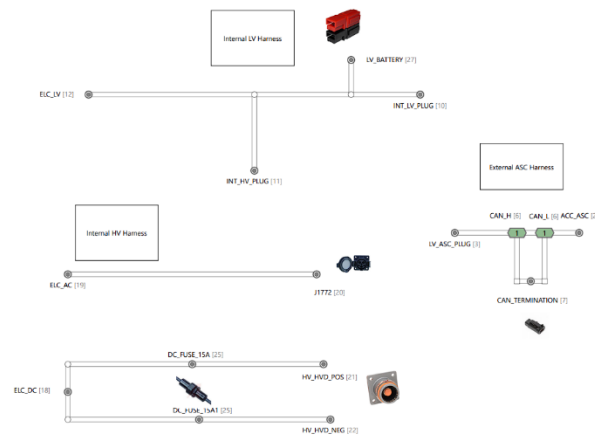


Figure 3: Top-level RapidHarness

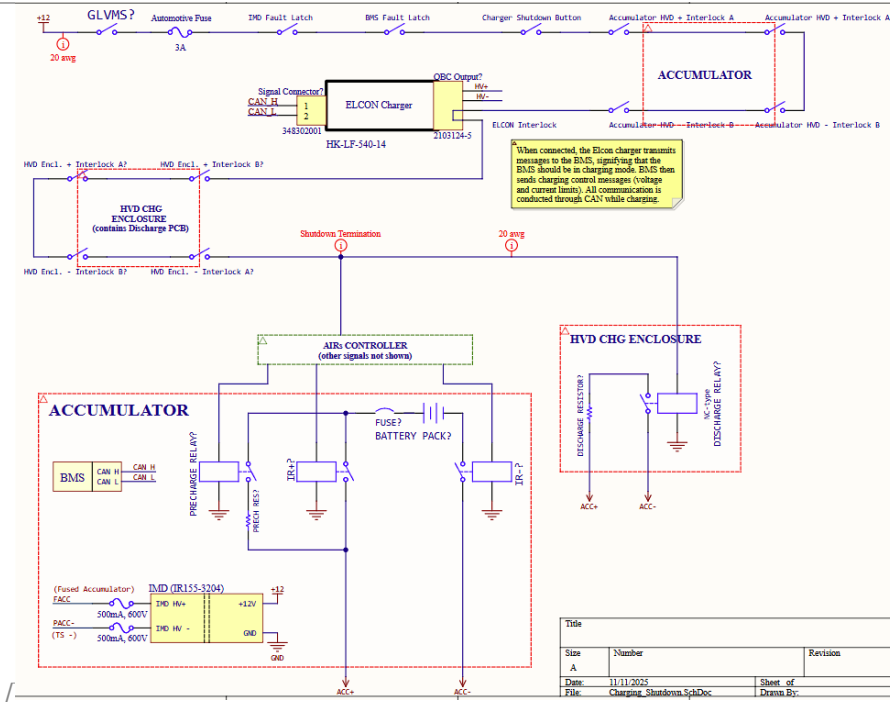


Figure 4: Charging shutdown circuit schematic overview

3. History of Past Designs

Cornell Racing's charging system has evolved significantly since the team's transition to an all-electric vehicle. Early FSAE electric vehicles commonly used commercial off-board chargers with liquid-cooled power stages to meet high charge-rate requirements [1].

3.1 BRUSA NLG513 Era

Cornell Racing originally used the BRUSA NLG513 charger. This unit had several significant limitations for competition use: it suffered from thermal management issues that caused charge interruptions, it could not reliably charge the accumulator to the full 600 V limit specified by EV.3.3.2, and it only supported Level 1 charging because it was not compatible with the J1772 protocol required for Level 2 EVSE [8]. These limitations made competition charging slow and unreliable.

3.2 Transition to Elcon Charger (ARG24)

The transition to the Elcon charger began in Spring 2024. The Elcon provided substantially better charging capacity and improved thermal behavior compared to the BRUSA system, and it is compatible with the J1772 interface through the AVC2.r controller [7]. Because the Elcon had already been validated in prior testing, ARG26 retained it as the core power conversion component.

3.3 ARG25 Charger

For ARG25, the team continued using the Elcon charger and consolidated the charging system into a portable case with reduced complexity [7]. The ARG25 design was functionally capable but suffered from several significant problems. The system was maintained by multiple contributors without centralized documentation, making it difficult to understand, debug, and verify against the rulebook. The most consequential rule-compliance gap was that the TSMPs were not shrouded, which caused the accumulator to fail a section of Electrical Technical Inspection at competition. This experience demonstrated that minor non-compliant details can produce major delays at competition.

Additional ARG25 issues included incomplete labeling, harnessing that was difficult to trace, and vehicle shutdown board logic being reused in the charger without removing vehicle-specific elements. These problems directly informed every major ARG26 design decision.



Figure 5: ARG26 charger packaging

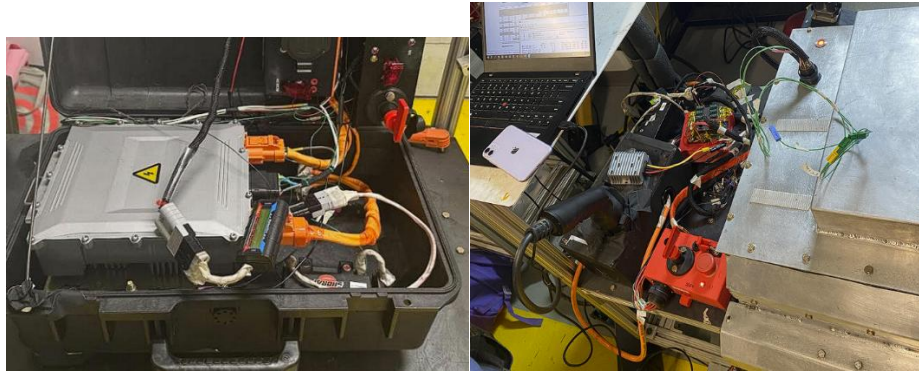


Figure 6: ARG25 and ARG24 chargers

4. ARG26 Design

4.1 Overall Design Approach

The ARG26 charging system was designed around the proven Elcon charger as its core power conversion component. Rather than redesigning power electronics, effort was focused on the surrounding system: the case, enclosures, shutdown PCB, and harnessing. Each enclosure was independently connectorized so it can be removed, inspected, or replaced without disturbing other subsystems. This is critical for serviceability because charger failures are often intermittent and difficult to localize without the ability to isolate subsystems.

The charging case was redesigned so that the Elcon sits in a taller vertical orientation, producing a case that is longer but narrower than ARG25. This orientation improves access to the Elcon output connections and allows the HV enclosure to be mounted on the side of the case while

keeping all major components accessible. The case mounts to the side of the charging cart using aluminum extrusion brackets, allowing it to remain on the cart during competition while still being usable flat on a bench during testing.

System

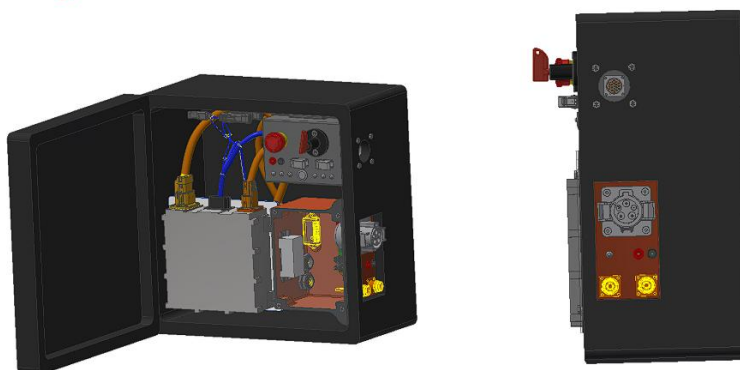


Figure 7: ARG26 charger case CAD — overall packaging

4.2 Design Changes Since Fall 2025 Technical Report

Two significant design changes were made after the Fall 2025 technical report [5]. The first was a rewire of the shutdown circuit to include a secondary switch controlling the SST. This switch replaced the one previously used to enable the Elcon's low-voltage power input. The rationale was that once the Elcon's internal LV rail becomes available, that higher-voltage source will naturally dominate the current draw through the LV diode network, making a separate Elcon LV power switch redundant. The secondary switch instead functions similarly to a TSMS in the vehicle shutdown circuit: it prevents the charging shutdown circuit from closing automatically as soon as LV power is applied, making charger startup a deliberate two-step process.

The second change was to disconnect the HV enclosure indicator LED. During integration testing, the AVC2.r output was found to be incapable of reliably sinking the current required to drive the LED, which degraded the AVC2.r output voltage and reduced system reliability. The LED was disconnected for ARG26. Future designs requiring indicators driven from AVC2.r outputs should buffer those outputs through a transistor or small driver IC rather than driving LEDs directly.

4.3 High-Voltage Enclosure

The HV enclosure is an ASA 3D-printed housing mounted on the side of the charging case. It contains the charger output connections from the Elcon, an in-line UL Class CC fuse with a serviceable screw-cap holder, the discharge board, shrouded TSMPs, SurLok Plus high-voltage connectors, and the AVC2.r interface connections. All internal high-voltage conductors are orange, consistent with EV.8.2.5.

The discharge board is a fixed-resistor network activated by the shutdown circuit. When the shutdown circuit opens, the discharge path closes and begins dissipating stored charge. The resistor values were selected to bring the output from 600 V to below 60 V in under five seconds into the expected capacitive load, satisfying EV.8.4.1. The discharge board was verified functionally prior to integration.

SurLok Plus connectors were selected for the charger-to-accumulator interface to match the connector type used on the accumulator. A critical assembly detail was polarity keying: the charger-side receptacles and plugs had to be oriented as mirror images of the accumulator side to produce correct polarity at the mating interface. Because the accumulator-side connectors had swapped polarity relative to the datasheet orientation, the charger-side leads were also swapped accordingly [4].



Figure 9: HV enclosure — CAD

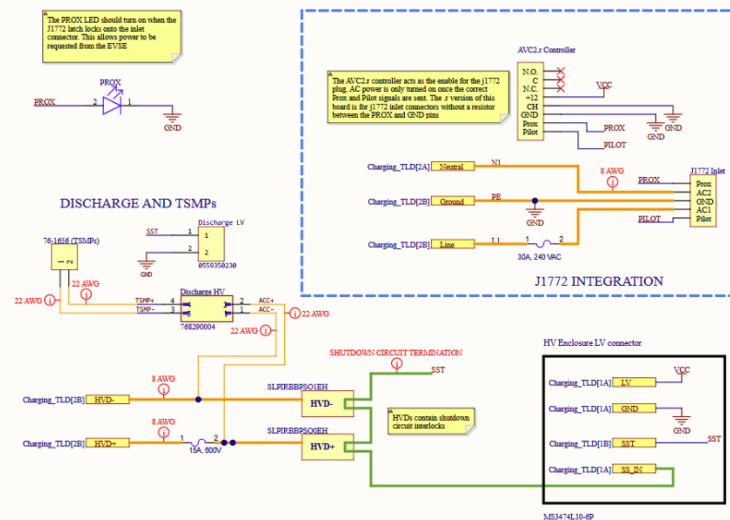


Figure 10: HV enclosure — schematic*Figure 11: HV enclosure — assembled photo*

4.4 Low-Voltage Control Enclosure

The LV enclosure is mounted on the top face of the charging case so that the charger shutdown button, reset button, and status indicators are all accessible during normal charging operation and during ETI, as required by EV.8.1.1 and EV.8.2.7. The enclosure uses mil-spec circular connectors for all harness interfaces so that it can be removed cleanly without cutting or depinning any wiring.

Inside the LV enclosure, components are mounted. The enclosure provides source-switching between the external LV battery and the Elcon-supplied 12 V rail. Isolation diodes on the shutdown PCB prevent the two sources from backfeeding into each other. Status indicator LEDs show which LV source is active, whether the charging shutdown circuit is closed, whether a fault is present, and whether HV is present on the charger output. LV measuring points allow direct multimeter access to the LV and ground reference nodes, which is useful when checking TSMP voltage or debugging shutdown circuit behavior.

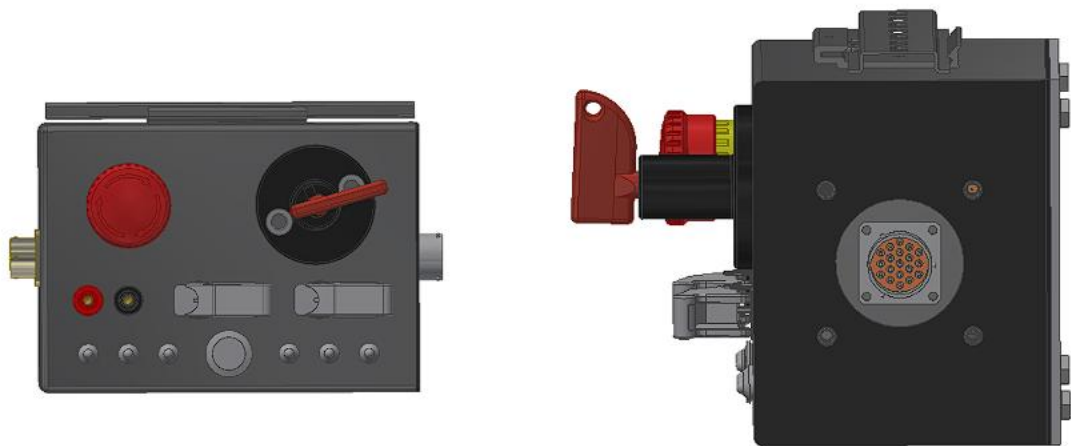


Figure 12: LV enclosure — CAD

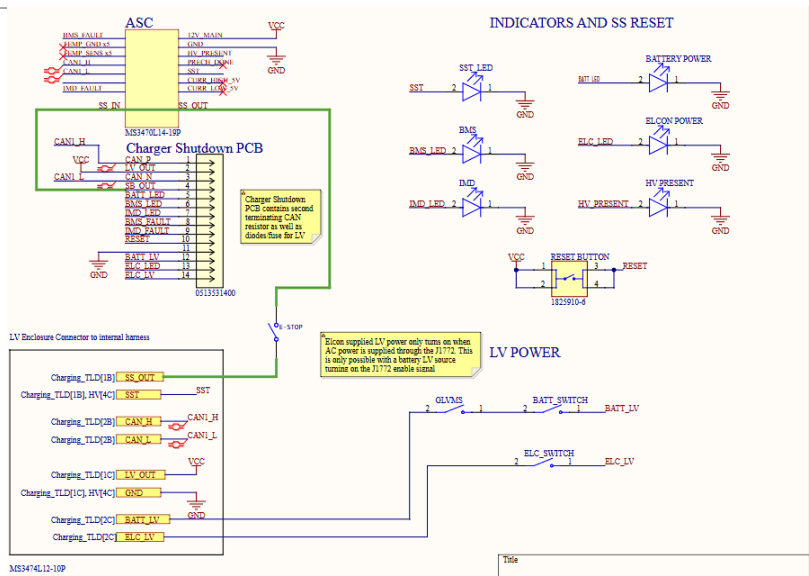


Figure 13: LV enclosure — schematic



Figure 14: LV enclosure — assembled photo

4.5 Charger Shutdown PCB

The charger shutdown PCB is a custom board designed specifically for the charging application. The ARG25 charger reused logic from the vehicle's shutdown board, which included vehicle-specific elements such as BSPD circuitry that are not required in a charger context. The ARG26 board removes all vehicle-only logic, reducing board area, simplifying the circuit, and eliminating potential confusion during debugging.

The board also consolidates three components that would otherwise require awkward placement in the harness: the two LV source isolation diodes, a 3 A LV fuse protecting the control circuit, and a 120 Ω CAN terminating resistor. Since the accumulator does not contain CAN termination, one terminating resistor is placed on the shutdown PCB and the other in the harness near the accumulator, placing the two terminators at opposite ends of the CAN bus.

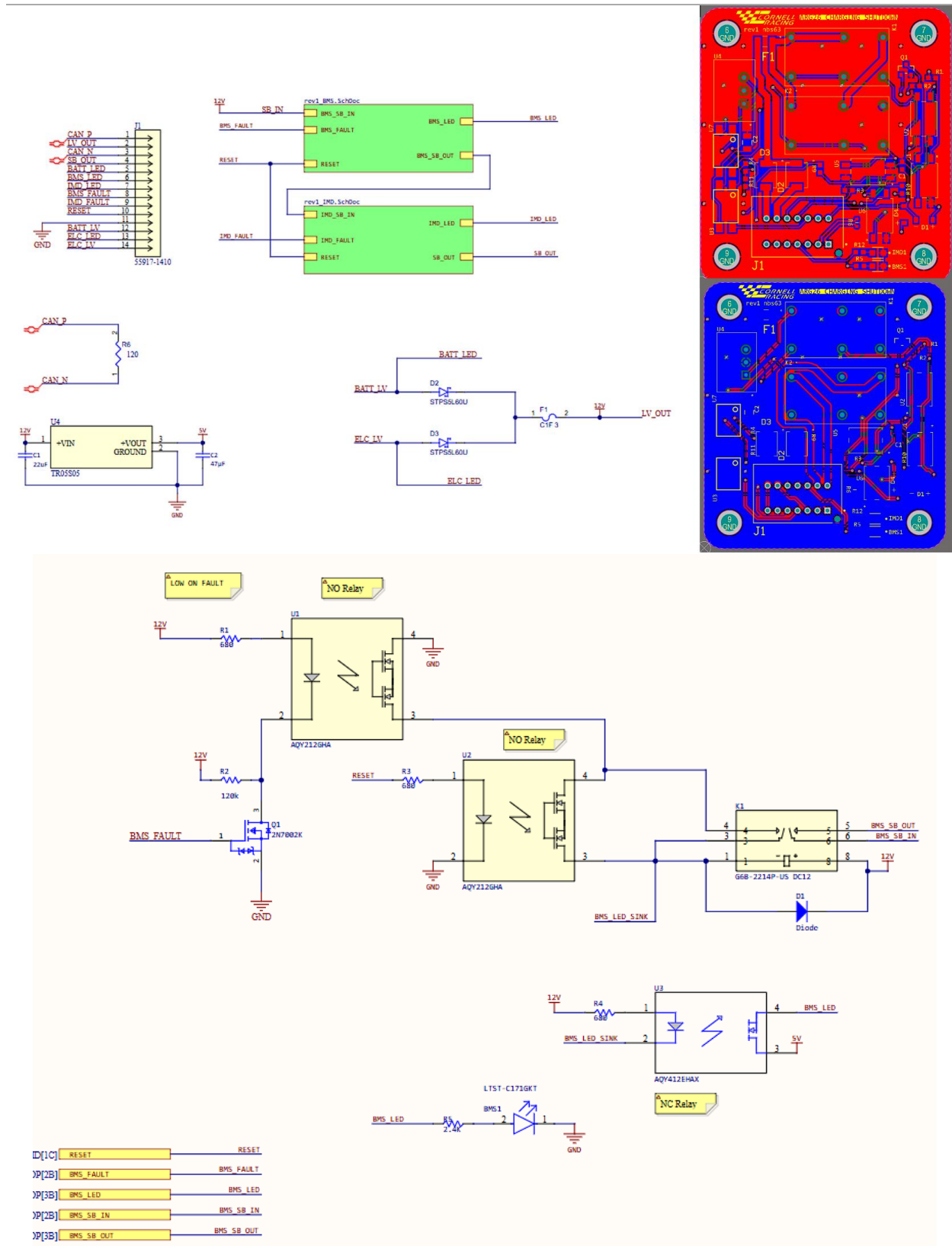


Figure 15: Charger shutdown PCB — schematic

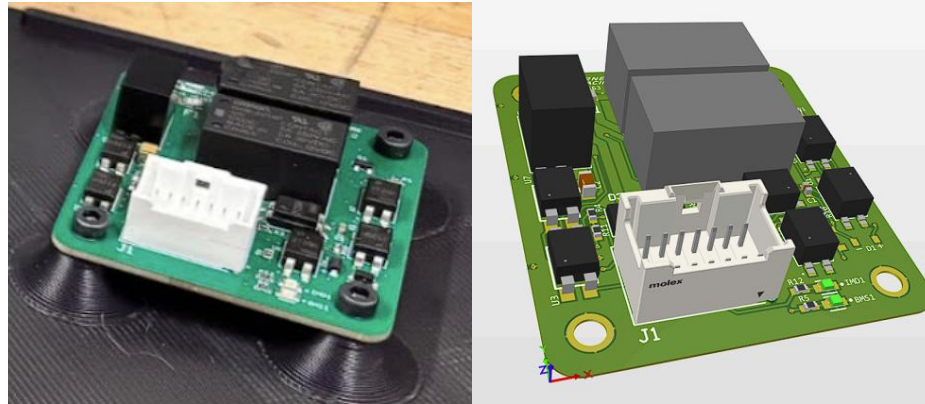


Figure 16: Charger shutdown PCB — assembled photo

4.6 Harnessing and Connectors

Harnessing was a high priority for ARG26 given the documentation and traceability problems encountered in ARG25. All system wiring was defined in RapidHarness and Altium schematics before assembly began, and a wire-numbering scheme was applied during assembly to maintain correct pinouts across all multi-pin connectors. Connector pinouts, wire gauges, color codes, and routing were captured in pinout tables stored on the S-drive alongside the design files.

The system uses mil-spec circular connectors for all LV enclosure interfaces and SurLok EMI Plus connectors for HV output connections. The LV connector selection allows the entire LV enclosure to be disconnected and removed for bench testing without accessing the interior of the case [4].

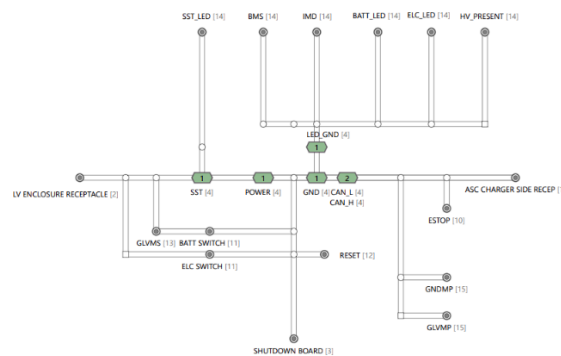


Figure 18: LV enclosure RapidHarness detail



Figure 19: Charger harness — assembled photo

4.7 Incomplete Aspects and Future Design Work

Several items remain incomplete or could be improved in future iterations. The HV enclosure indicator LED was disconnected due to AVC2.r drive limitations. Restoring a visual HV-present indicator would improve operator situational awareness during charging; the fix requires either a buffered LED driver or a separate LV supply circuit and does not require changes to the HV enclosure hardware.

The cart mounting was designed but not fully validated under dynamic transport conditions. The bracket attachment should be tested by rolling the cart over representative floor surfaces to confirm the charger does not shift or place strain on external harnessing. No detailed thermal model of the Elcon charger under sustained load was developed for ARG26. Prior testing showed the Elcon does not have major thermal issues [5].

5. Manufacturing and Assembly

Manufacturing the ARG26 charging system was substantially more time-intensive than anticipated. The system integrates many distinct component types — 3D-printed enclosures, mil-spec connectors, SurLok connectors, PCBs, fuse holders, DIN rail hardware, and external harnessing — each requiring different assembly techniques and tooling. Assembly was organized by subsystem: each enclosure was assembled, harnessed, and continuity-checked independently before being integrated into the full system.

5.1 Enclosure Manufacturing

The HV and LV enclosures were 3D-printed in ASA (acrylonitrile styrene acrylate). ASA was chosen over PLA because of its superior resistance to elevated temperature and UV exposure, both relevant in a competition environment where the charger may be used outdoors [5]. Heat-set threaded inserts were installed for all fastener interfaces. Fitment prototypes of both enclosures were printed before JanMan to verify component clearances and resolve layout conflicts before committing to final hardware.

5.2 High-Voltage Connector Assembly

The most significant manufacturing challenge was the SurLok Plus high-voltage connector assembly. Two issues arose. First, connector polarity keying required careful attention: the charger-side receptacles and plugs had to be oriented as mirror images of each other to produce correct polarity at the accumulator interface, and a polarity swap in the accumulator-side connectors required a corresponding swap on the charger side [4]. Second, the SurLok crimping dies available in the shop were found to be worn, producing inconsistent crimp geometry. Several crimps failed pull tests during assembly and had to be remade. For ARG27, crimp tooling condition should be verified and worn dies replaced before beginning HV connector manufacturing, and pull tests should be performed on each completed crimp.

5.3 Elcon Charger Mounting

The Elcon charger physical mounting features did not match the dimensions shown in the manufacturer datasheet [2]. Specifically, the mounting hole pattern differed from the drawing, requiring the designed mounting plate to be modified. A revised sheet-metal mounting solution was fabricated to accommodate the actual hardware. Future designs should measure the physical charger unit directly and verify all critical mounting dimensions before finalizing mounting plates or enclosure geometry.

5.4 ARG27 Manufacturing Recommendations

- Order all connectors, fuses, DIN rail hardware, and enclosure fasteners by November 2026. SurLok Plus connectors in particular have lead times that can delay assembly if ordered late.
- Verify condition of all crimp tooling before manufacturing begins. Replace worn SurLok crimping dies.
- Measure all critical dimensions on physical hardware (Elcon charger, AVC2.r, fuse holders) before finalizing enclosure CAD.
- Specify all intra-enclosure wire lengths in harness documentation prior to assembly.
- Schedule enclosure-level bring-up tests (continuity, PCB function) before full system integration.
- Store all design files (Altium, RapidHarness, CAD, PCB) on the S-drive with a consistent folder structure before the designer rolls off the project.

6. Test Results

Charger testing for ARG26 was primarily functionality-based. Because the Elcon charger itself had been validated in prior years, ARG26 testing focused on verifying that the surrounding system was assembled correctly, operated safely, and integrated properly with the accumulator.

6.1 Completed Testing

The following tests were completed:

- Enclosure-level continuity checks on both the HV and LV enclosures, verifying that all pinouts matched the Altium schematics and RapidHarness pinout tables.
- Charger shutdown PCB functional check, verifying correct switching behavior of the LV source isolation diodes, shutdown circuit relay, and CAN termination.
- Discharge board functional check, verifying that the discharge path activated correctly when the shutdown circuit input was removed.
- Full-system operation check across startup and shutdown sequences, verifying LV source switching, shutdown circuit closure, and charger enable/disable behavior.
- Accumulator charging: the charger successfully charged with the accumulator connected, confirming that the system performs its intended function.

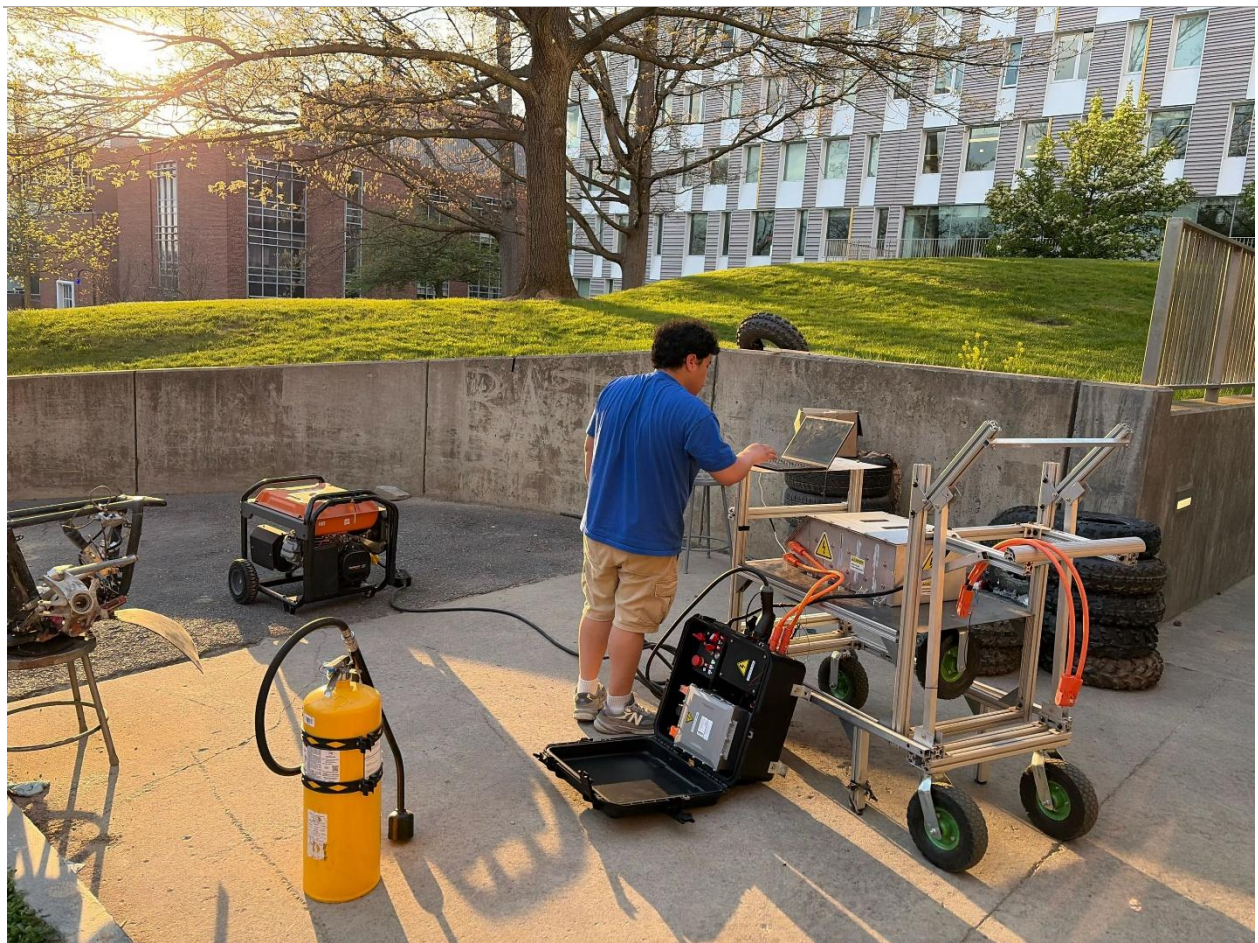


Figure 24: Jacob Goatzonlez charging the pack with a generator

6.2 Data Collection

No formal quantitative test report was produced for the charging system. The primary outputs of the testing process are Altium schematics, RapidHarness pinout tables, enclosure CAD files, PCB design files, assembly photos, and notes from bring-up testing, all stored on the S-drive

under Powertrain Peripherals / Charging System [9][10][11]. Functional outcomes were confirmed but not captured in structured data logs.

6.3 Future Testing

Because the charger is functional, future testing should focus on maintaining reliability rather than re-validating the design.

- Normal charge verification: confirm the charger completes a regular charge cycle without interruption, and check that the Elcon, enclosure surfaces, and connectors do not reach temperatures suggesting thermal stress.
- Shutdown circuit check: press the charger shutdown button and confirm that current to the accumulator stops immediately. Trigger the BMS and IMD shutdown paths independently and confirm equivalent behavior.
- Cart mounting inspection: confirm the charger remains securely attached to the cart and no external harnessing is under tension after transport over typical floor surfaces.
- Harness and connector inspection: confirm all SurLok connectors are fully seated and latched, all mil-spec connector shells are tightened, and no orange HV wire insulation shows damage or abrasion.

7. References

- [1] Formula SAE, “Formula SAE Rules 2026,” SAE International, Warrendale, PA, 2026.
- [2] Elcon / TC Charger, “Elcon Charger User Manual and Datasheet,” TC Charger Corp., manufacturer documentation, 2023.
- [3] Modular EV Power, “AVC2.r Active Vehicle Controller — Technical Reference,” Modular EV Power, manufacturer documentation, 2024.
- [4] Amphenol Industrial, “SurLok Plus / SurLok EMI Plus High-Voltage Connector Datasheet,” Amphenol Industrial Operations, Sidney, NY.
- [5] N. Su, “ARG26 Fall Technical Report: Charging System,” Cornell Racing, Powertrain Peripherals, Ithaca, NY, Dec. 2025.
- [6] N. Su, “ARG26 Spring Midsemester Technical Report: Charging System,” Cornell Racing, Powertrain Peripherals, Ithaca, NY, Apr. 2026.
- [7] Cornell Racing, “ARG25 Technical Report: Charging System / Cell Tab Research,” Powertrain Peripherals, Ithaca, NY, Spring 2025.
- [8] Cornell Racing, “ARG25 Fall Technical Report: Charging,” Powertrain Peripherals, Ithaca, NY, Fall 2024.
- [9] Cornell Racing, “ARG26 Charging System Altium Schematics and PCB Files,” S:\Reports\2026 Car\Spring Technical\Powertrain Peripherals\Charging System.
- [10] Cornell Racing, “ARG26 Charging System RapidHarness Files and Pinout Tables,” S:\Reports\2026 Car\Spring Technical\Powertrain Peripherals\Charging System.
- [11] Cornell Racing, “ARG26 Charging System CAD and Assembly Photos,” S:\Reports\2026 Car\Spring Technical\Powertrain Peripherals\Charging System.